The Social Cost of Carbon — ENV462 Exercise

Name: _____ Roll number: _____

> No discussion, phones or talking is allowed during the activity. Time: 10 minutes.

What does a cost-benefit IAM do?

Cutting emissions costs money and emitting causes damage. A cost-benefit IAM looks for the level of emissions where the total of these two costs is as small as possible. That happens where the cost of cutting one more tonne equals the damage that tonne would cause, which is the crossing point on the graph. The price at that point is the social cost of carbon, and a carbon tax set at that level would deliver the optimal emissions. Models such as DICE do exactly this, only over many years and with a climate model linking emissions to temperature and damages.

Exercise

The world emits 50 $GtCO_2$ per year if nothing is done. Find the emissions level and carbon price a cost-benefit IAM would recommend.

Emissions (Gt)	**30**	**35**	**40**	**45**	**50**
Cost of cutting one more tonne ($/t)	80	60	40	20	0
Damage from one more tonne ($/t)	30	35	40	45	50

1. Sketch both curves with emissions on the horizontal axis and find where they cross. What are the optimal emissions and carbon price?

2. Suppose damages turn out to be twice as large. Does the carbon price go up or down, and roughly to what?

The Social Cost of Carbon — ENGG562 Exercise

Name: _____ Roll number: _____

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What does a cost-benefit IAM actually do?

Cutting emissions costs money and emitting causes damage. A cost-benefit IAM looks for the level of emissions where the total of these two costs is as small as possible. That happens where the cost of cutting one more tonne equals the damage that tonne would cause, which is the crossing point on the graph. The price at that point is the social cost of carbon, and a carbon tax set at that level would deliver the optimal emissions. Models such as DICE do exactly this, only over many years and with a climate model linking emissions to temperature and damages.

Exercise

The world emits 50 $GtCO_2$ per year if nothing is done. Find the emissions level and carbon price a cost-benefit IAM would recommend.

Let E be global emissions in $GtCO_2$ per year. With no action, $E = 50$. When emissions are E, the cost of cutting one more tonne is $4(50 - E)$ dollars, and the damage from one more tonne is E dollars.

1. Find the emissions level where the two are equal. What are the optimal emissions and carbon price?

2. Suppose damages turn out to be twice as large, so the damage from one more tonne is $2E$ dollars. Find the new optimal emissions and carbon price. Did the carbon price double? Explain why or why not.